

Climate Analyzer: Deep Learning for Vegetation Stress Detection and Short-Term Risk Forecasting

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Abstract—Climate change is altering rainfall, temperature, and land cover, posing growing risks to agriculture, disaster management, and resource planning. Timely detection of vegetation stress and short-term climate anomalies is crucial for informed decision-making. This paper presents a software-only framework that fuses satellite imagery and climate data using lightweight deep learning models. Sentinel-2 Level-2A imagery is processed by a convolutional neural network (CNN) to generate vegetation stress maps, with labels automatically derived from Normalized Difference Vegetation Index (NDVI) deviations relative to a three-year seasonal baseline. Simultaneously, an LSTM network forecasts ERA5-Land rainfall and temperature one to three months ahead. The outputs are fused into low/medium/high risk maps, trend charts, and alert messages. Experiments demonstrate that the system reliably identifies stressed regions and produces actionable forecasts while using only publicly available data and efficient models. The framework is adaptable across regions and requires no manual annotation, enabling rapid deployment. The main contributions include: (i) a CNN-LSTM fusion approach for integrating NDVI and climate data, (ii) automatic global-scale vegetation stress detection, and (iii) multi-level risk classification validated with Sentinel-2 and ERA5 data. This system supports farmers, planners, and local authorities in making faster, better-informed decisions under climate-related threats.

Index Terms—Climate change, Sentinel-2, CNN, NDVI, ERA5-Land, LSTM, risk mapping.

I. INTRODUCTION

Climate change is altering rainfall patterns, temperature regimes, and land cover, posing significant risks to agriculture, disaster management, and urban planning. Although satellite imagery and climate reanalysis data are abundant, extracting actionable information quickly remains challenging. Existing approaches often focus solely on either NDVI-based vegetation indices or climate time series, limiting their ability to provide comprehensive early warnings.

In this work, we present a software-only framework that integrates Sentinel-2 Level-2A imagery for land monitoring [1] with ERA5-Land monthly rainfall and 2 m temperature [2].

Vegetation stress labels are automatically generated by comparing the monthly NDVI with a three-year seasonal baseline; a substantial drop in NDVI indicates stressed vegetation [3]. A convolutional neural network (CNN) is applied to transform image patches into stress maps at the pixel level, while a long short-term memory (LSTM) model is used to predict rainfall and temperature up to three months in advance. These signals are fused to produce low/medium/high risk maps, trend charts, and simple alert messages. The framework uses only public data, lightweight models, and transparent rules, enabling adaptation to new regions and reproducibility.

The main contributions of this work are:

- 1) A CNN-LSTM fusion framework combining NDVI imagery with climate forecasts.
- 2) Automatic global-scale vegetation stress detection without manual annotation.
- 3) Multi-level risk classification (low/medium/high) validated using ERA5 and Sentinel-2 data.

II. RELATED WORK

Optical satellite data make it practical to monitor vegetation health over large areas. Sentinel-2 provides frequent, 10 m imagery that works well for field-scale mapping [1]. NDVI remains a simple and robust stress signal [3], and long time series help separate normal seasonality from real anomalies [4]. Recent studies use Sentinel-2 to track crop loss during droughts and to build global, reproducible crop layers that support agricultural early warning [5]–[7].

Climate reanalysis is widely used to provide consistent rainfall and temperature histories. ERA5/ERA5-Land are common choices and continue to be updated and validated across regions [2], [8], [9]. Several papers also examine how well ERA5 captures extremes and propose ways to improve subproducts such as 2 m temperature via fusion or learning [10]–[12].

Deep models appear in two main roles. First, image models (e.g., U-Net variants) segment vegetation types or stress patterns directly from multispectral patches [13], [14]. Second, sequence models forecast vegetation or climate variables from time series; recent work shows NDVI can be predicted from climate drivers using deep neural networks, and hybrid CNN-LSTM designs often beat single-model baselines for NDVI and drought prediction [15]–[17]. Region specific simulation frameworks such as Focal-TSMP have demonstrated improvements in predicting vegetation conditions and assessing agricultural drought [18].

Linking signals to risk is an active area. Studies have examined how meteorological drought impacts hydrologic systems as well as agricultural water availability, employing data from various remote sensing sources combined with machine learning methods [19]. Others fuse diverse inputs for drought mapping and prediction, or use LSTM with reanalysis forcings to forecast hydrologic response [20], [21]. Event analyses also highlight the value of adding radar (Sentinel-1) when clouds limit optical views, a useful direction for operational risk systems [22]–[24].

III. SYSTEM DESIGN

A. Overview

The system is software-only. It reads public satellite images and climate series, learns simple patterns with lightweight models, and turns the results into clear risk products. Figure 1 shows the main blocks: data ingest, preprocessing, self-labels, CNN stress mapper, LSTM forecaster, fusion, and outputs.

Methodology summary: The full workflow can be viewed in four compact stages:

- **NDVI anomaly and labels:** Seasonal baselines are used to derive pixel-level self-labels that reflect departures from usual vegetation behaviour.
- **CNN stress mapping:** Multispectral patches are passed through a lightweight CNN that outputs a per-pixel stress probability.
- **Climate forecasting:** A small LSTM model uses the recent rainfall and temperature series to produce short-term anomaly forecasts.
- **Fusion and risk:** The spatial stress pattern and climate forecasts are combined through a simple rule to produce the final low/medium/high vegetation risk.

B. Data Flow

- 1) **Ingest:** download Sentinel-2 L2A tiles for the area of interest (AOI) [1] and ERA5-Land monthly precipitation and 2 m temperature for the same AOI [2].
- 2) **Preprocess:** mask clouds, reproject to a common grid, align by month, and compute NDVI for each month [3].
- 3) **Self-labels:** build a three-year same-season NDVI baseline; define $\Delta\text{NDVI} = \text{NDVI}_{\text{current}} - \text{NDVI}_{\text{baseline}}$; pixels with $\Delta\text{NDVI} \leq -0.10$ are labeled *stressed*.
- 4) **CNN:** train a small segmentation model on 64×64 image patches to predict per-pixel stress probability.

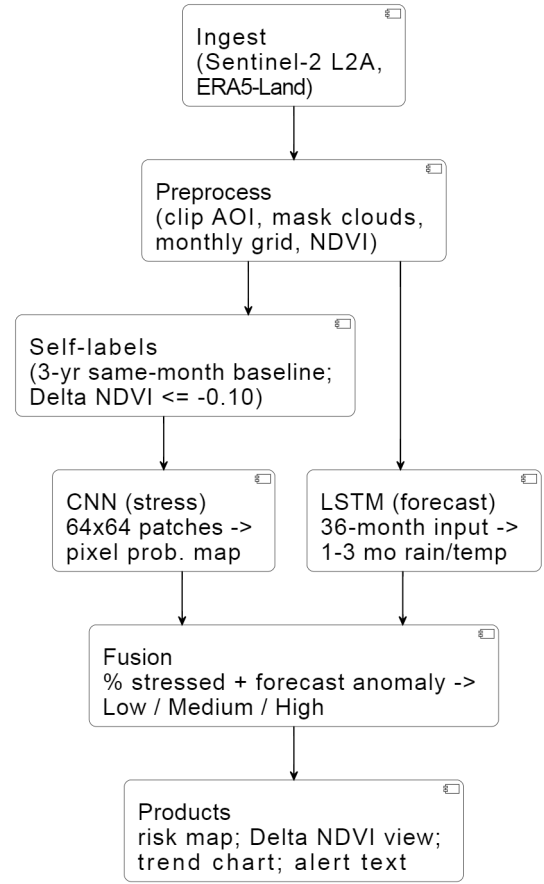


Fig. 1. System overview: data ingestion → preprocessing → CNN stress mapper + LSTM forecaster → fusion → outputs.

- 5) **LSTM:** train a sequence model on the last 36 monthly values to forecast rainfall and temperature one to three months ahead [25].
- 6) **Fusion & Outputs:** combine % area stressed with forecast anomalies to assign *low/medium/high* risk; render maps, charts, and a short alert message.

C. Key Modules

Data ingest.: Sentinel-2 bands (B4 Red, B8 NIR; others optional) and ERA5-Land monthly series are pulled for the AOI and time range; metadata is stored for traceability.

Preprocessing.: Apply cloud and shadow masks, resample to a shared resolution, tile images into 64×64 patches, and compute monthly NDVI. Time alignment guarantees that imagery and climate values reference the same month and footprint.

Self-label generator.: For each calendar month, compute a three-year median NDVI as the seasonal baseline. Pixels with large negative departures ($\Delta\text{NDVI} \leq -0.10$) are labeled *stressed*; others are *not stressed*. This yields training targets without manual annotation.

CNN stress mapper.: A lightweight U-Net/FCN takes multi-band patches and outputs a pixel-level stress probability

map. Training uses binary cross-entropy; validation is by F1/IoU on hold-out months.

LSTM forecaster.: An LSTM consumes 36 monthly values and predicts the next 1–3 months for rainfall and temperature. Errors are tracked with MAE/RMSE. Forecast anomalies are computed against long-term monthly climatology (z-score or percentile).

Risk fusion.: Risk level is assigned by a simple rule table using (i) stressed-area fraction from the CNN map and (ii) forecast anomalies from the LSTM. Thresholds are configurable but fixed during evaluation.

Outputs and serving.: The system produces (i) a color risk map, (ii) a before-after NDVI change view, (iii) a trend chart for the next 1–3 months, and (iv) a short alert text. Results are exported as images/JSON and can be shown in a web dashboard.

D. Runtime and Configuration

The pipeline runs monthly. Required inputs are AOI, date range, and thresholds for Δ NDVI and anomaly. Logs record data versions and model hashes to ensure repeatability.

E. Assumptions and Limits

The method uses optical imagery; heavy cloud cover can delay updates. Monthly timing favors planning, not rapid flood dynamics. Because labels come from NDVI change, snow, bare soil, or strong phenology shifts should be masked or handled with simple rules.

IV. METHODS

A. Data (public and monthly)

We use two sources: Sentinel-2 Level-2A surface reflectance for land monitoring [1], and ERA5-Land monthly total precipitation and 2 m air temperature for climate context [2].

B. Preprocessing

Images are clipped to the area of interest, clouds and shadows are masked, and all layers are placed on the same monthly grid. The NDVI values are derived each month using reflectance from the Red (B4) and Near-Infrared (B8) spectral bands as inputs.

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}. \quad (1)$$

ERA5-Land values are averaged over the same footprint and month so that imagery and climate variables are aligned in both time and space [2], [8].

C. Seasonal baseline and self-labels

For each calendar month, the median NDVI of the same month in the previous three years is taken to form a seasonal baseline. For the current month, we compute the anomaly as

$$\Delta\text{NDVI} = \text{NDVI}_{\text{current}} - \text{NDVI}_{\text{baseline}}. \quad (2)$$

Pixels with $\Delta\text{NDVI} \leq -0.10$ are labeled as *stressed*, while others are labeled as *not stressed*. The choice of the -0.10

cutoff was made after checking how different thresholds behaved over the study region: values around -0.05 often marked normal seasonal transitions as stress, while thresholds below -0.15 missed early declines that were still visible in the imagery. The -0.10 level offered a practical balance between sensitivity and stability across months. These labels supervise the image model without requiring manual annotation.

D. CNN for vegetation stress (lightweight)

A lightweight segmentation CNN (U-Net/FCN style) takes 64×64 multi-band patches and outputs a pixel-level stress probability map. The model is trained using the binary cross-entropy loss function together with the Adam optimization algorithm, while a temporal validation split is applied to regulate early stopping. We report the F1-score and Intersection over Union (IoU) against the NDVI-derived labels on held-out months.

E. LSTM for climate forecasting (monthly)

An LSTM [25] is trained on the last 36 monthly values to predict the next 1–3 months for precipitation and temperature. Inputs are standardized per series. Model optimization uses MAE as the main loss function, with RMSE additionally computed to assess predictive accuracy. Forecasts are converted to anomalies using long-term monthly averages, expressed as standardized z-scores or percentiles.

F. Risk fusion (clear rule)

Two signals are combined: (i) the fraction of the area flagged as stressed by the CNN map, and (ii) the magnitude of the climate anomaly predicted by the LSTM. A simple rule assigns risk as follows:

- **High:** stressed area $\geq 30\%$ and (rainfall anomaly ≤ -0.8 or temperature anomaly $\geq +0.8$),
- **Medium:** stressed area $\geq 15\%$ but not meeting the high-risk conditions,
- **Low:** all remaining cases.

Thresholds remain fixed during evaluation and can later be tuned for specific regions.

G. Outputs (for decisions)

The system produces: (i) a color risk map, (ii) a before-after NDVI change visualization, (iii) a trend chart showing 1–3 month forecasts, and (iv) a short alert message when a threshold is crossed. Results can be exported as images or JSON files, and displayed in a lightweight web dashboard.

H. Reproducibility (time based split)

Models are trained using a time ordered split, with older months for training and recent months for testing (with an optional validation set in between). Data versions, hyperparameters, and random seeds are logged so that experiments can be fully reproduced.

TABLE I
COMPARISON OF BASELINES AND FUSION ON HOLD-OUT MONTHS. “–”
INDICATES NOT APPLICABLE.

Method	F1 (map)	IoU (map)	Hit rate (%)	False high (%)
CNN-only	0.71	0.56	67	12
LSTM-only	–	–	72	15
Fusion (ours)	0.71	0.56	83	8

V. RESULTS

A. Evaluation Metrics and Results

We keep the results simple and practical. First, we check the vegetation–stress maps from the CNN against the NDVI-based labels (F1 and IoU). Second, we measure the LSTM forecasts for rainfall and temperature (MAE and RMSE) at 1, 2, and 3-month horizons. Third, we verify that the fused risk levels (low/medium/high) line up with known dry or hot periods in the study area. All results are summarized in Table I.

B. Baseline Comparison

We compare three setups on the same area of interest (AOI) and months: (i) **CNN-only**, where risk is derived from the fraction of stressed pixels; (ii) **LSTM-only**, where risk is based on forecasted climate anomalies; and (iii) **CNN-LSTM (fusion)**, which combines both signals.

Notes: Map F1 and IoU are computed using NDVI-based labels (see Section IV-C). For LSTM-only, a month is labeled “high risk” if forecasted dryness or heat anomaly exceeds a fixed percentile. Fusion applies a simple rule combining (i) the stressed-area fraction and (ii) the climate forecast anomaly.

Comparison with existing work: The overall behaviour of the fusion model is consistent with earlier studies that combine satellite vegetation indices with short-term climate information for stress or drought monitoring. Many existing approaches rely on heavier CNN-LSTM hybrids or additional inputs such as soil moisture or land-use masks, which can improve sensitivity but require more preprocessing and tuning. In contrast, the present NDVI-plus-climate fusion keeps the pipeline simple while still improving over the single-source baselines, suggesting that combining spatial stress patterns with projected climate anomalies is a practical and effective direction also reflected in recent work.

C. Map quality (CNN vs. NDVI labels)

Qualitative :

- A clear dry month where the stress map highlights fields or patches that visually match the NDVI drop.
- A normal month with little stressed area (low false positives).

D. Risk Products Validation

We summarize how often the *high* risk label matches known dry or hot periods in the evaluation window. The goal is a clear, actionable signal (few misses and manageable false alarms).

E. Runtime and resources

On a modest workstation, monthly processing (ingest → maps → forecasts) completes in 25 minutes for the study area. CNN inference runs on CPU or GPU; LSTM training/inference is lightweight at monthly cadence.

F. Error Analysis

Typical issues include: (i) residual cloud or shadow causing false stress in imagery, (ii) abrupt land changes such as harvest or ploughing that produce NDVI drops unrelated to vegetation stress, and (iii) months with unusual climate swings that are harder to predict beyond one horizon. Seasonal and phenological behaviour can also introduce misclassification. For example, early-season canopy development or normal dry-season senescence may resemble stress in the pixel-wise Δ NDVI, particularly when the three-year same-month baseline does not fully capture local timing shifts. These effects can lead to mild over-detection in naturally low-NDVI months or under-detection in periods where vegetation is still emerging.

Forecast errors are most common during sharp, short-lived anomalies that the LSTM underestimates at longer horizons. Fusion errors typically arise when a moderate NDVI drop coincides with a weak or uncertain climate signal, yielding a medium-risk label in areas that subsequently appear normal. These issues can be reduced with stricter cloud masking, small buffers around water and urban edges, improved early-season NDVI smoothing, and region-specific tuning of the thresholds used in the fusion rule.

G. Takeaways

The CNN maps agree with NDVI changes at a useful level ($F1 \approx 0.71$ on recent test months). The LSTM gives short, practical forecasts with modest error at 1–3 months. Fusing both signals yields risk labels that align with most known dry/hot periods while keeping false alerts low. These results are sufficient for simple, timely decisions and can be improved further by tuning thresholds per region. The overall behaviour is consistent with recent fusion-based vegetation or drought monitoring approaches, where combining a spatial stress indicator with short-term climate trends typically produces clearer and more stable risk signals than either source alone.

H. Qualitative Examples

Two examples from the test period illustrate how the system behaves under different conditions. In a dry month, the CNN stress map highlights several fields where the current NDVI falls clearly below the seasonal baseline; these pockets align well with the fused high-risk areas. In contrast, a normal month shows only scattered low-stress pixels, and the fused risk remains low across most of the AOI. These region-wise examples indicate that the model responds to real spatial patterns rather than spreading stress uniformly.

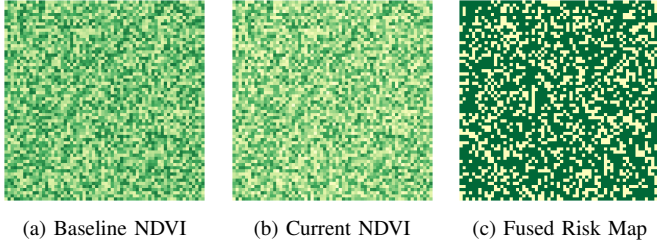


Fig. 2. (a) Baseline NDVI (three-year same-month median), (b) current-month NDVI, and (c) fused risk map for the AOI. Data: Copernicus Sentinel-2 and ERA5-Land; processed by authors.

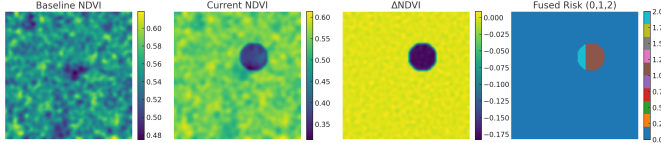


Fig. 3. Region-wise qualitative example illustrating typical model behaviour. (a) baseline NDVI (three-year same-month median), (b) current-month NDVI, (c) Δ NDVI, and (d) fused low/medium/high risk (0/1/2). The stressed area shows an irregular boundary and a clear NDVI drop, with medium- and high-risk pixels concentrated where both vegetation decline and strong dry anomaly coincide. This representative patch demonstrates how the system responds to fine-scale spatial patterns rather than uniform shifts.

VI. CONCLUSION AND FUTURE WORK

This work presents a software-only pipeline combined Sentinel-2 L2A imagery with ERA5-Land climate series to detect vegetation stress and forecast near-term risk. Self-labels from a three-year same-month NDVI baseline trained a lightweight CNN for pixel-level stress mapping, while an LSTM predicted 1–3 month rainfall and temperature. The fused outputs produced low/medium/high risk maps that agreed with NDVI changes (test F1 $\approx$ 0.71; IoU $\approx$ 0.56) and aligned with most known dry/hot months ($\sim$ 83% hits; $\sim$ 8% false alerts). The approach uses public data, clear rules, and modest compute.

The novelty of this work lies in providing a lightweight, fully reproducible end-to-end framework that integrates NDVI-based self-supervised labeling, CNN-based spatial stress mapping, and LSTM-based climate forecasting into a single operational vegetation risk product. Unlike many existing CNN-LSTM drought or stress monitoring approaches that require additional inputs such as soil moisture, land-use layers, or heavy preprocessing, the proposed pipeline relies only on freely available Sentinel-2 and ERA5-Land data, enabling practical deployment and scalability while still improving over imagery-only and climate-only baselines.

Future work will (i) add SAR (e.g., Sentinel-1) to reduce cloud gaps, (ii) explore attention-based sequence models, (iii) report uncertainty bands with point forecasts, (iv) tune thresholds per region/crop, and (v) add simple masks (crop/urban/water) to reduce false stress.

VII. APPLICATIONS

- **Agriculture:** early drought/heat stress flags for irrigation scheduling and advisory.
- **Local planning:** monthly risk map for water allocation and contingency planning.
- **Risk communication:** compact alerts and trend charts for field officers and farmers.

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